Benders Decomposition. Small Examples

Large Scale Optimization

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1. Small Examples of MILP

A very simple example

$$\left.\begin{array}{ll} z = \min\limits_{x} & x \\ s.t: & x \in [0,2] \; or \\ & x \in [3,4] \end{array}\right\} \longrightarrow z^* = 0 \; and \; x^* = 0$$

$$\begin{array}{rll|l} v(\delta) = \min\limits_{x,\delta} & x & & D.V. \\ s.t: & x \leq 2 + M(1-\delta) & & u_1^- \\ & x \geq -M(1-\delta) & & u_1^+ \\ & x \leq 4 + M\delta & & u_2^- \\ & x \geq 3 - M\delta & & u_2^+ \\ & \delta \in \{0,1\} & & \end{array}$$

Formulations. MILP

$$
\begin{array}{l|l}
\phi^* = & \\
= \min\limits_{x,\delta} \quad x^+ - x^- & D.V. \\
s.t. \quad \begin{pmatrix} 1 & -1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x^+ \\ x^- \\ \sigma_1^+ \\ \sigma_1^- \\ \sigma_2^+ \\ \sigma_2^- \end{pmatrix} = \begin{pmatrix} 2+M \\ -M \\ 4 \\ 3 \end{pmatrix} - \delta \begin{pmatrix} M \\ -M \\ -M \\ M \end{pmatrix} & \begin{matrix} u_1^- \\ u_1^+ \\ u_2^- \\ u_2^+ \end{matrix} \\
x^+, x^-, \sigma_1^+, \sigma_1^-, \sigma_2^+, \sigma_2^- \geq 0, \quad \delta \in \{0,1\} &
\end{array}
$$

Reference MILP formulation

$$
\begin{array}{lll}
\phi^* = \mathsf{Min}_{x,y} & f^\top y + c^\top x & \\
\text{s.t.} & Dx = d - Fy & \mid \; u \\
\text{(MILP)} & x \geq 0, y \in Y &
\end{array}
$$

Formulations. Subproblem in Primal form

$$
\begin{array}{lll|l}
v(y) = & & & \\
= \min\limits_{x} & x^+ - x^- & & D.V. \\
 & & & \\
 & s.t. & \begin{pmatrix} 1 & -1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x^+ \\ x^- \\ \sigma_1^+ \\ \sigma_1^- \\ \sigma_2^+ \\ \sigma_2^- \end{pmatrix} = \begin{pmatrix} 2+M \\ -M \\ 4 \\ 3 \end{pmatrix} - \delta \begin{pmatrix} M \\ -M \\ -M \\ M \end{pmatrix} & \begin{array}{c} u_1^- \\ u_1^+ \\ u_2^- \\ u_2^+ \end{array} \\
 & & x^+, x^-, \sigma_1^+, \sigma_1^-, \sigma_2^+, \sigma_2^- \geq 0 &
\end{array}
$$

Linear problems for each $y \in Y$

$$
\begin{array}{lll}
v(y) = \mathsf{Min}_x & c^\top x \\
\qquad\qquad \text{s.t.} & Dx = d - Fy \\
\text{(P)} & x \geq 0
\end{array}
$$

A very simple example. Subproblem in Dual form

$$
\begin{aligned}
v(\delta) = \max_{u} \quad & (2+M-\delta M)u_1^- + M(\delta-1)u_1^+ + (4+\delta M)u_2^- + (3-\delta M)u_2^+ \\
s.t. \quad & \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} u_1^- \\ u_1^+ \\ u_2^- \\ u_2^+ \end{pmatrix} \leq \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
v(\delta) = \max_{u} \quad & (2+M-\delta M)u_1^- + M(\delta-1)u_1^+ + (4+\delta M)u_2^- + (3-\delta M)u_2^+ \\
s.t. \quad & u_1^- + u_1^+ + u_2^- + u_2^+ = 1 \qquad \hat{u}^0 = (0,1,0,0)^\top, \quad \vec{w}^0 = (-1,1,0,0)^\top \\
& u_1^+, u_2^+ \geq 0, u_1^-, u_2^- \leq 0 \qquad \hat{u}^1 = (0,0,0,1)^\top, \quad \vec{w}^1 = (0,0,-1,1)^\top
\end{aligned}
$$

Linear problems for each $y \in Y$

$$
\begin{aligned}
v(y) = \text{Max}_u \quad & (d - Fy)^\top u \\
\text{(D)} \qquad \text{s.t.} \quad & D^\top u \leq c
\end{aligned}
$$

Possible solutions

Case δ =0.

$$v(0) = \min_x \quad x$$
$$s.t: \quad -M \leq x \leq 2 + M$$
$$3 \leq x \leq 4$$

$$Primal\ solution\ x^* = 3$$

Dual solution (by complementary slackness):
$u_1^- = u_1^+ = u_2^- = 0 \longrightarrow \hat{u}^0 = (0, 0, 0, 1)^\top$

Case δ =1.

$$v(1) = \min_x \quad x$$
$$s.t: \quad 0 \leq x \leq 2$$
$$3 - M \leq x \leq 4 + M$$

$$Primal\ solution\ x^* = 0$$

Dual solution (by complementary slackness):
$u_1^- = u_2^+ = u_2^- = 0 \longrightarrow \hat{u}^1 = (0, 1, 0, 0)^\top$

Notice that for any value of δ, the subproblem is always feasible: $Y_0 = Y$

Optimality cuts:

For a given dual extreme vector $\hat{u}$ of the dual polyhedron, the general optimality cut of the MASTER PROBLEM will be:

$$z \geq 0 \cdot \delta + \left(\begin{pmatrix} 2+M \\ -M \\ 4 \\ 3 \end{pmatrix} - \delta \begin{pmatrix} M \\ -M \\ -M \\ M \end{pmatrix} \right)^{\top} \begin{pmatrix} \hat{u}_1^- \\ \hat{u}_1^+ \\ \hat{u}_2^- \\ \hat{u}_2^+ \end{pmatrix}$$

or, in scalar form:

$$z \geq 0 \cdot \delta + (2 + M - \delta M)\hat{u}_1^- + M(\delta - 1)\hat{u}_1^+ + (4 + \delta M)\hat{u}_2^- + (3 - \delta M)\hat{u}_2^+$$

A checking based on linear programming duality:

Sometimes $\hat{u}$ is obtained indirectly (i.e., when subproblems are solved using primal forms).
Assume that the subproblem has been solved for $\bar{y}$ and that a finite feasible subproblem is obtained.

- If $\hat{u}$ is the extreme vector obtained from the binding variables set at $\bar{y}$, then:

$$\phi(\bar{y}) = f^\top \bar{y} + (d - F\bar{y})^\top \hat{u}$$

For our small example:
- $\delta = 0, \hat{u}^0 = (0,0,0,1)^\top$

$$\phi(0) = 3 = 0 \cdot \cancel{\delta} + (2+M)\cancel{\hat{u}_1^-} + M\cancel{\hat{u}_1^+} + 4\cancel{\hat{u}_2^-} + 3\hat{u}_2^+ = 3$$

- $\delta = 1, \hat{u}^1 = (0,1,0,0)^\top$

$$\phi(1) = 0 = 0 \cdot \delta + 2\cancel{\hat{u}_1^-} + 0\hat{u}_1^+ + (4+M)\cancel{\hat{u}_2^-} + (3-M)\cancel{\hat{u}_2^+} = 0$$

Application of the Benders decomposition

- Iteration 0. Take some initial guess, say $\delta = 0$. Set $\varphi^0 = -M'$

- Iteration 1. Solve the associated Subproblem for $\delta = 0$
  Solution: Feasible Subproblem : $v(0) = 3$, $\hat{u}^0 = (0,0,0,1)^\top$.
  Because $\varphi^0 = -M'$, relgap $= \dfrac{3 + M'}{M' + \varepsilon} \approx 1 + 3/M' > \epsilon$
  For $\hat{u}^0 = (0,0,0,1)^\top$, the corresponding feasibility cut is:

$$z \geq 3 - M\delta$$

. * Master Problem:

$$\begin{aligned} \varphi^{(1} = \min_{z,\delta} \quad & z \\ & z \geq 3 - M\delta \\ & \delta \in \{0,1\} \end{aligned} \quad \longrightarrow \ \delta^* = 1,\ \varphi^{(1} = 3 - M$$

Application of the Benders decomposition

Iteration 2. Solve the Subproblem for $\delta = 1$
- Solution: $v(1) = 0,\ x^* = 0,\ \hat{u}^1 = (0, 1, 0, 0)^\top$
Optimality check: $v(1) = 0 > \varphi^{(1} = 3 - M$: positive gap.
Notice that $\delta = 1$ does not verify the optimality test at this iteration. Benders Decomposition does not identify it (at this stage) as the solution of the problem.
Reason: the master problem does not have enough cuts.
For $\hat{u}^1 = (0, 1, 0, 0)^\top$ the optimality cut is: $z \geq M(\delta - 1)$.
- The new cut is added to the Master problem:

$$\varphi^{(2} = \min_{z,\delta} \quad z$$
$$\begin{aligned} & z \geq 3 - M\delta \\ & z \geq M(\delta - 1) \\ & \delta \in \{0, 1\} \end{aligned} \qquad \longrightarrow \ \delta^* = 1,\ \varphi^{(2} = 0$$

Application of the Benders decomposition

- Iteration 3. For $\delta = 1$ the subproblem is solved (again*).
$\Rightarrow v(1) = 0$.
- Optimality check: relgap $= (0 - 0)/(0 + \varepsilon) < \epsilon$. STOP:

$$\delta = 1,\ x^* = 0 \text{ is the solution.}$$

*Notice that in two subsequent iterations the master problem has provided the same value of the binding variables $\bar{y}$. This may only happen if actually $\bar{y}$ is the solution of the problem. Otherwise, an implementation error is present and simply this would lead to the algorithm to stagnate at $\bar{y}$.

Another simple example

$$\left.\begin{array}{rl} z = \min\limits_{x} & x \\ s.t: & x \in [0,2] \; or \\ & x \in [3,4] \; or \\ & x \in [6,7] \\ & x \leq 5 \end{array}\right\} \longrightarrow z^* = 0 \; and \; x^* = 0$$

$$\begin{array}{rlll} v(\delta) = & \min\limits_{x,\delta} & x & \\ s.t: & u_1^+ & -M(1-\delta_1) \leq x \leq 2 + M(1-\delta_1) & u_1^- \\ & u_2^+ & 3 - M(1-\delta_2) \leq x \leq 4 + M(1-\delta_2) & u_2^- \\ & u_3^+ & 6 - M(1-\delta_3) \leq x \leq 7 + M(1-\delta_3) & u_3^- \\ & & x \leq 5 & u_4^- \\ & \text{(P)} & \delta_1 + \delta_2 + \delta_3 = 1 & \\ & & \delta \in \{0,1\}^3 & \end{array}$$

Another simple example

The primal problem is infeasible for $\delta_3 = 1,\ \delta_2 = \delta_1 = 0$

$$
\begin{aligned}
v(\delta) = \max_{u} \quad & (2 + M(1-\delta_1))u_1^- + (0 - M(1-\delta_2))u_1^+ + \\
& (4 + M(1-\delta_2))u_2^- + (3 - M(1-\delta_2))u_2^+ + \\
& (7 + M(1-\delta_3))u_3^- + (6 - M(1-\delta_3))u_3^+ + 5u_4^- \\
s.t. \quad & u_1^- + u_1^+ + u_2^- + u_2^+ + u_3^- + u_3^+ + u_4^- = 1 \\
\text{(D)} \quad & u_i^+ \geq 0,\ u_i^- \leq 0, i = 1,2,3,\ \ u_4^- \leq 0
\end{aligned}
$$

Another simple example

The dual problem is unbounded for $\delta_3 = 1$, $\delta_2 = \delta_1 = 0$.
To see it, the previous dual problem must be set in standard form ($\max \longrightarrow \min$; define new variables $v_i^- = -u_i^-$, $v_i^+ = u_i^+$
Simplex tableau for the dual in standard form:

v_1^-	v_1^+	v_2^-	v_2^+	v_3^-	v_3^+	v_4^-	
-1	1	-1	1	-1	1	-1	1
$(2+M)$	M	$4+M$	$M-3$	7	-6	5	0
$M-4$	$M+6$	$M-2$	$M+3$	1	0	-1	6

This shows that an extreme direction for the variables v is $\vec{v} = (0,0,0,0,0,1,1)^\top$ and for the variables u, $\vec{u} = (0,0,0,0,0,1,-1)^\top$. Notice that if $c(\delta)$ is the cost vector of the dual problem (D), then

$$c(\delta)^\top \vec{u} = \begin{cases} > 0 & if\ \delta_3 = 1 \\ < 0 & if\ \delta_3 = 0 \end{cases}$$

Application of the Benders decomposition

- Iteration 0. Take some initial guess, say $\delta_1 = \delta_2 = 0, \delta_3 = 1$. Set $\varphi^0 = -M'$

- Iteration 1. Solve the associated Subproblem for $\delta = 0$
Solution: Infeasible Subproblem. The extreme direction of unboundedness, $\vec{u} = (0,0,0,0,0,1,-1)^\top$, for the dual problem (D) is obtained.
- Master Problem:

$$\varphi^{(1} = \min_{z,\delta} \quad z \qquad\qquad z^{(1} = \text{undefined}$$

$$\begin{array}{l} 0 \geq 1 - M(1-\delta_3) \\ \delta_1 + \delta_2 + \delta_3 = 1 \\ \delta \in \{0,1\}^3 \end{array} \longrightarrow \delta_3^* = 0, \quad \begin{array}{l} \delta_1 = 1,\ \delta_2 = 0 \\ or \\ \delta_1 = 0,\ \delta_2 = 1 \end{array}$$

Either of the solutions $(1, 0, 0)$ or $(0, 1, 0)$ can be used for the subproblem of the next iteration. Arbitrarily, $\delta = (1, 0, 0)^\top$ is chosen.
The constraint $0 \geq 1 - M(1 - \delta_3)$ will always guarantee that $\delta_3 = 0$

$$Y_0 = \{\delta \in \{0, 1\}^3 |\ \delta_1 + \delta_2 = 1,\ \delta_3 = 0\}$$

- Iteration 2. Subproblem for $\delta = (1,0,0)^\top$. Optimal solution found. An extreme vertex fo the dual polyhedron is obtained $\hat{u} = (0,1,0,0,0,0,0)$, $v(0,1,0,0,0,0,0) = 0$.

$$\text{gaprel} > \epsilon$$

- Master problem:

$$\begin{aligned} \varphi^{(2} = \min_{z,\delta} \quad & z \\ & 0 \geq 1 - M(1-\delta_3) \\ & z \geq M(\delta_1 - 1) \\ & \delta_1 + \delta_2 + \delta_3 = 1 \\ & \delta \in \{0,1\}^3 \end{aligned}$$

$$z^{(2} = -M,\ \delta_1^* = 0,\ \delta_2^* = 1,\ \delta_3^* = 0,$$

- Iteration 3. Finite Subproblem for $\delta = (0,1,0)^\top$. An extreme vertex fo the dual polyhedron is obtained $\hat{u} = (0,0,0,1,0,0,0)$, $v(0,1,0) = 3$.

$$\text{gaprel} = \frac{3+M}{M+\varepsilon} \approx 1 + \frac{3}{M} > \epsilon$$

- Master problem:

$$\begin{aligned} \varphi^{(3} = \min_{z,\delta} \quad & z \\ & 0 \geq 1 - M(1-\delta_3) \\ & z \geq M(\delta_1 - 1) \\ & z \geq 3 + M(\delta_2 - 1) \\ & \delta_1 + \delta_2 + \delta_3 = 1 \\ & \delta \in \{0,1\}^3 \end{aligned}$$

$$z^{(3} = 0,\ \delta_1^* = 1,\ \delta_2^* = 0,\ \delta_3^* = 0$$

- <u>Iteration 4</u>. Finite Subproblem for $\delta = (1, 0, 0)^\top$. An extreme vertex fo the dual polyhedron is obtained $\hat{u} = (0, 1, 0, 0, 0, 0, 0)$, $v(1, 0, 0) = 0$.

$$\text{gaprel} = 0. \text{ STOP!}$$

Solution of the problem: $\delta = (1, 0, 0)^\top$